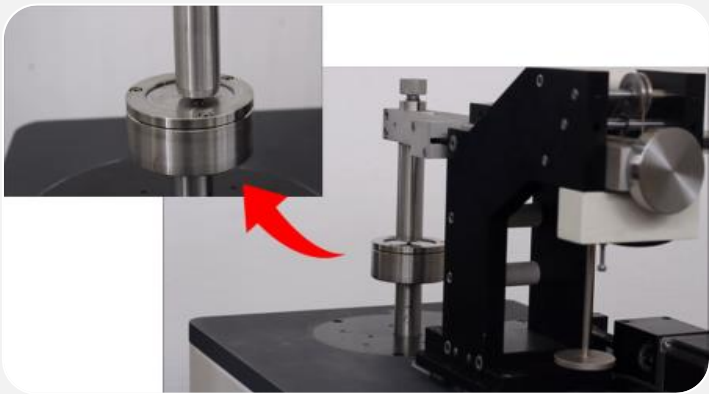


Research Facilities in the Institute

S. No.	Name of Lab/ Equipment	Description/ Applications
1	Nvidia DGX 2 server(5 petaFlops system) An Artificial Intelligence (AI) Lab with high-performance Graphics Processing Units (GPU) server	This artificial intelligence lab is one of the few state-of-the-art labs in India and first in Uttar Pradesh, equipped with NVIDIA-DGX-2 which is the world's first 2 petaflops system, packing the power of 16 of the world's most advanced GPUs and accelerating the newest deep learning model types that were previously untrainable. With ground-breaking GPU scale, models can be trained 4X bigger on a single node. In comparison with legacy x86 architectures, DGX-2's ability to train ResNet-50 would require the equivalent of 300 servers with dual Intel Xeon Gold CPUs. The Artificial Intelligence Lab can also help to train students, developers, data scientists, and researchers to use deep learning and accelerated computing to solve real-world problems across a wide range of domains. With access to GPU-accelerated workstations in the cloud, researchers and students will learn how to train, optimize, and deploy neural networks using the latest deep learning tools, frameworks, and SDKs. They can also learn how to assess, parallelize, optimize, and deploy GPU-accelerated computing applications. GPU computing leverages the parallel processing capabilities of GPU accelerators and enabling software to deliver dramatic increase in performance for scientific, data analytics, engineering, consumer, and enterprise applications. 
2	BET Measurement System	The equipment is useful for determining the surface area of materials. Potential application includes all those areas where the high surface area of materials is a necessity. 

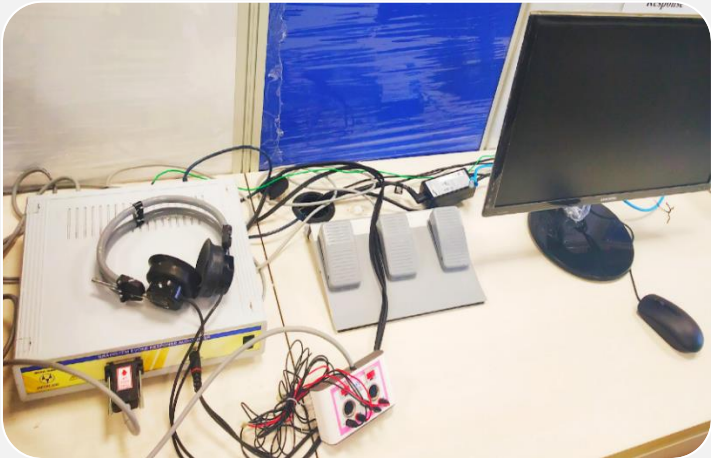
3	Industrial Robotic Center	An advanced state of the art Industrial Robotic Centre is established in the institute by KUKA India Private Limited (A German Company). The robotic centre has three robotic cells, Material handling Cell, MIG welding Cell, Bare robotic cell. The KUKA robots facility is used to understand and train basic concepts of robot programming, motion planning, and programming methodology of robotic systems, performing research-oriented specific tasks including pick and place job. MIG welding cell is the Bare robot which students mainly use to learn, implement and test their new ideas, methods, and techniques using robotic systems. The purpose of Bare robotic cell is to use the arm for the testing and analysis of the systems developed by the students. The robotic manipulator can be customized as per the systems designed and developed by the students. 
4	Industrial Automation Lab	Industrial Automation Lab offers manufacturing automation on Flexible Manufacturing Systems (FMS) including material storing and dispatching equipment AS/RS, machining stations CNC lathe, CNC milling, CNC grinding, material handling equipment such as 4 axis and 6 Axis Robot, loading /unloading automated guided vehicle(AGV), Assembly Station and Visual Inspection Setup. This Lab provides in-depth coverage of Computer Integrated Manufacturing and Industrial Automation. It provides hands-on study particularly in the areas of computer Aided Manufacturing (CAM), Computer Numerical Control (CNC) and manufacturing automation. It has tool dynamometer (Kistler company) and minimum quantity lubrication setup along with Temperature Data Acquisition system. The speciality of this Lab is that the CNC machines installed here can be individually utilized for research purpose. 

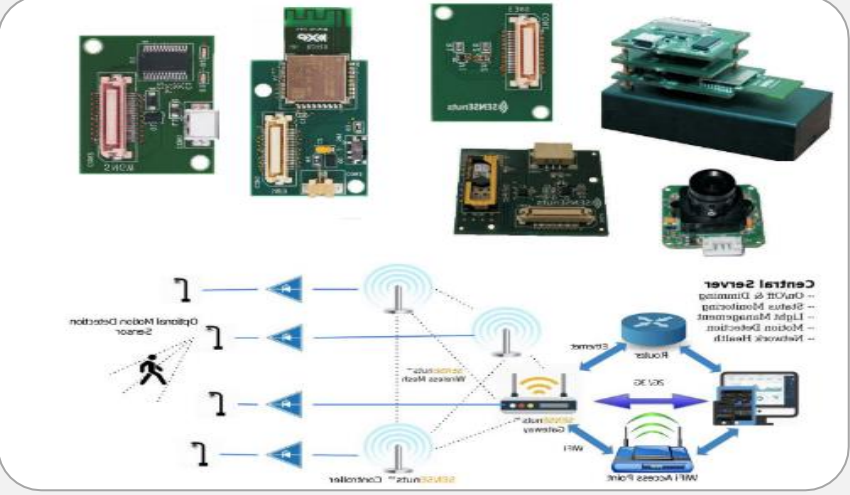
5	3D Printing Lab	A method of manufacturing is known as ‘Additive manufacturing’, due to the fact that instead of removing material to create a part, the process adds material in successive patterns to create the desired shape. 3D Printing uses software that slices the 3D model into layers (0.01mm thick or less in most cases). Each layer is then traced onto the build plate by the printer, once the pattern is completed, the build plate is lowered and the next layer is added on top of the previous one. Main areas of use are: Prototyping, Specialized parts – aerospace, military, biomedical engineering, dental, Hobbies and home use, Future applications– medical (body parts), buildings and cars. The entire 3D printing technology can be divided into 3 steps – (a) 3D Design (b) Slicing (c) 3D Printing. Two types of printers are available in our center. These 3D printers work on the principal of fused material deposition and filament-based polymers deposition to shape an object, layer-by-layer. 
6	Design and Simulation Center	This lab consists of 18 high end specifications computer systems (32 GB high speed RAM, 4 GB dedicated Graphics card and high-performance multicore Intel processor) with latest design and simulation software like SolidWorks, ANSYS, and MATLAB etc., which enables the students to simulate various complex designs. This lab is integrated with 3D printing lab so as to design and develop the prototypes for testing. Using the facility and MATLAB, the students can develop innovative solutions to challenging research problems. A few application areas are Antenna Design and Placement, Autonomous Sensor Development, Avionics and Flight Control, Electric Motors, Electronics Reliability, E-Motor NVH and Active Sound Design for EV, Gas Turbines, Hypersonics, PCBs, ICs and IC Packages, Power Electronics, Rotating Machinery, Vehicle Chassis, Wind Turbine Design. 

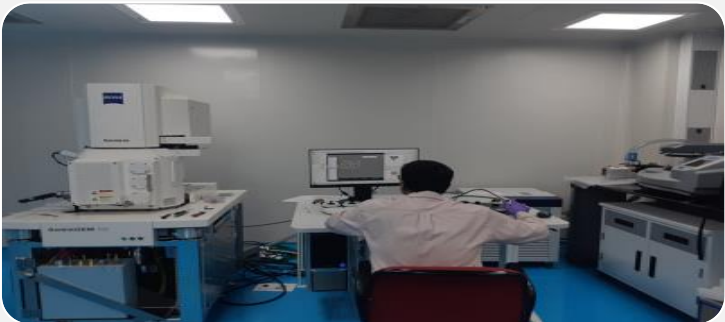
7	Sensor Drives & Control Lab	This is a fundamental lab for students from different backgrounds to make them understand the working and usage of different types of sensors, controllers and drives. Students are taught about the working principle, sensors usability in embedded systems, actuators usability in automation and embedded systems. They are also trained to develop their own sensors/ actuation system, according to the project requirements. Starting from semiconductor, Diode, transistors, Logic Gate, the students are trained to program different microcontrollers using different high level language like- C++, Python, Matlab etc. so that they can have the capacity and competence to develop different smart system of their own for the service and benefit of human society. 
8	Pin on disc tribometer	A pin on disc tribometer consists of a stationary pin that is normally loaded against a rotating disc. The pin can have any shape to simulate a specific contact, but cylindrical tips are often used to simplify the contact geometry. This Ducom Pin/Ball on Disk Tribometer is a test instrument designed for accurate and repeatable tribological characterization of bulk materials, coatings and lubricants. The Pin/Ball on Disk Tribometer is used to characterize the tribological behavior for almost all material bench-marking and materials development needs. Different types of objectives can be resolved related to analysis of wear characteristics of industrial materials, composites, metallic, ceramic components with different lubricant and varying loads. 

9	Human Non-invasive Blood Pressure (NIBP) Monitor	Reliably record and monitor trends in response to interventions on finger arterial pressure, systolic, diastolic, mean arterial, heart rate and interbeat interval. Equipped with Volume Clamp Technology that provides accurate finger pressure measurement and Height Correction unit which can automatically correct movement of the finger cuff relative to the heart, (HCU) Heart Rate Variability HRV (psychophysiology) Biopotentials & GSR (psychophysiology) Record and measure various physiological responses such as biopotential recordings (EOG, ECG, EMG, EEG, GSR). In Cardiovascular studies, ECG signals are recorded to examine heart rate, heart rate variability, and analysis of the waveform morphology, arrhythmia and other similar functions, and sleep studies. 
10	Equi-vital Physiological Monitoring System	Comfortable belts for recording physiological data via a compact and unobtrusive sensor belt, leaving the subject free to move naturally. Module can record ECG signal, Acceleration (3 axis), Respiration, Skin Temperature, SPO2 and GSR. The system should provide real time data recording with online and offline analysis. Provide Noise and movement artifactcillary pack. It is used to record and measure various physiological responses such as EOG, ECG, EMG, EEG and Galvanic Skin Response (GSR). 

11	Marker less Motion capture system (2-D Gait Analysis System) with Force plate	It contains 16 * 2 feet stable three layered platform. The software provides parameters like velocity, cadence, step length, step width, stance and swing percent, toe in/out angle, instantaneous. Centre of Pressure (COP), Centre of Mass, Foot angle, direction of progression, Left to Right Ratios path efficiency, total pressure, footfall COP, and [COP-COMe] etc. There is also a facility for multiple protocols like walking, standing, running and jumping. It can also be used for comparing variables between the left and right footfalls to evaluate symmetry, track changes overtime to quantify patient progress, and provide metrics and complete reports. Video synchronization of 2 or more cameras with replay capabilities and a push-button switch for marking events, is also available. It is used in research and study of Cerebral Palsy, Stroke, Parkinson's Disease, Multiple Sclerosis, Alzheimer's Disease, Ataxia, Muscular Dystrophy, Neuropathy Concussion, Fall Risk, Osteoarthritis, Motor Impairments, Sensory Impairments, Knee and Hip Replacement, Injury Rehabilitation, Amputation, Brace fitting and tuning, Prosthetic limb optimization, Equines Gait (Toe-Walking), Spine Bifida. 
12	EMOTIV EPOC+ 14 Channel Mobile EEG	EMOTIV EPOC+ 14 channel EEG provides access to professional grade brain data. Access high-quality raw EEG data, conduct research leveraging our detections for mental commands, analyse performance metrics or facial expressions. The EPOC+ measures both EEG and 9-axis motion data. Data is transmitted wirelessly through Bluetooth. It is a 5-channel wireless EEG device covering frontal, temporal and parietal and occipital locations around the brain. It utilizes proprietary polymer sensors that are hydrophilic they pull moisture from the air and skin. EMOTIV EPOC+ 14 channel mobile EEG is designed for scalable and contextual human brain research and advanced brain-computer interface applications. 

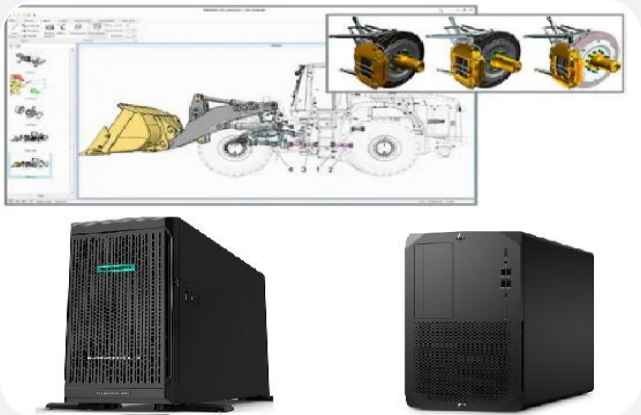
13	Auditory brainstem response (ABR) audiometry system	Auditory brainstem response (ABR) audiometry is a neurologic test of auditory brainstem function in response to auditory (click) stimuli, which is the most common application of auditory evoked responses. ABR audiometry refers to an evoked potential generated by a brief click or tone pip transmitted from an acoustic transducer in the form of an insert earphone or headphone. The elicited waveform response is measured by surface electrodes typically placed at the vertex of the scalp and ear lobes. It features high performance amplifiers specifically designed to obtain high quality evoked potentials signals. With 14 bit averages, the BERAGRAPH can measure an objective auditory threshold with accuracy greater than 5 db and distinguish end cochlear deafness from retrocochlear deafness. The auditory stimulator gives burst, login and click stimulation presented either in condensation, refraction or alternating from 0 db to 120 db. The masking intensity can be controlled automatically with the stimulating intensity while maintaining the same programmed difference. 
14	Delsys Wireless EMG System	Freely record and measure electrical activity produced by muscles, using wireless EMG sensors. Wireless EMG is useful for a wide range of applications within exercise physiology like track & field and gymnastics, studying motor control for patient rehabilitation or muscle performance, activity, and fatigue in elite athletes, measure electrical muscle activity. 

15	FLIR E5 Thermal Imaging Camera	This is a real-time working 10,800 (120 × 90) pixel infrared resolution, focus-free camera. Provides thermal images along with accurate temperature measurements. The FLIR E5 is a powerful thermal imaging tool for troubleshooting electrical, mechanical, and building problems. Find hidden issues quickly, easily, and safely—without the need for direct contact with equipment. 
16	SENSEnats	SENSEnats uniquely offers full technology stack with application layer, wireless network & cloud connectivity protocols for rapid IoT prototyping and also end to end vertical applications. It is Microcontroller with integrated 802.15.4 transceiver and variety of sensors like Environment, Meteorological, Air & water quality etc. It also has features like modular design having Gateways, Radios & sensors devices. It is easy to use because of C based programming, exhaustive set of “easy to use” APIs, flexible MAC protocol implementation, live data Interface with MATLAB etc. It has many advanced features to offer like energy efficient individual StreetLight Controllers which enable remote On/Off switching, dimming control, User configurable time scheduling & grouping schemes, Current/power consumption tracking of luminaire, Alerts for outage & malfunction, connectivity to Local or Cloud Server for data access and management and interfaces for Sensors like Motion detection, Pollution, light etc. 

17	Material Chemistry & Synthesis Laboratory	The laboratory is equipped with all the basic and modern infrastructural facilities like fume hood, spot extractors, magnetic stirrers, ultrasonic processor, hot air ovens, distilled water setup, microwave, vacuum furnace, for enabling academic and research activities. The preparation of a variety of nanomaterials for gas-sensing, materials for battery and super capacitors, can be achieved here. 
18	Field Emission Scanning Electron Microscope (FE-SEM) with EDAX module	The equipment is highly useful for viewing the shape, size, texture and elemental composition of materials at micro and nano scale those that cannot be observed by the unaided eye. One of the most important characterization technique for studying, the morphology of materials at the micro/nano scale, EDAX is used for determining the elemental composition of the materials. 
19	Electrochemical Workstation	This instrument is used for carrying out electrochemical energy storage, heavy metal ion detection, and catalysis related researches. Application areas include battery, supercapacitors, corrosion, etc. 

20	Impedance Analyser	Useful for studying the electrical impedance in materials. Characterization of fuel cells, photovoltaic cells, batteries and other electrochemical devices. 
21	Voyager Cyber city Simulator Hardware Modules For Cyber Simulator (smart TV LED, Dell Power Edge R230 with 19.5 screen Dell optiplex 7050 with 19.5 screen)	Cyber city simulator at Centre for Advanced studies is the facility for automated modelling and monitoring of cyber security threats at smart city environments. It displays status for cyber security threats on roads, power station, airport, metro rail, automobiles etc. It has state of the art tools and equipment for smart city projection. The objectives of this lab are to design and create attack plan methodologies, to understand social engineering aspects used for attacks, to get an insight into enterprise security trend, to use latest techniques to hack into systems and networks, to conduct regular audits and penetration test in your company, to support legal team with Digital forensic evidence, to support compliance roadmaps based on standards for your organization, to support Internal Audit teams for IT security compliance. 

22	Solar Simulator	Solar simulator is a device for simulating natural sunlight in laboratory conditions. Used for the preparation and testing of solar cells and modules. 
23	Hp z238 Microtower Workstations in Google Developers Code	This lab is constructed with an objective to replicate coding experience at Google. The specifications of the installed computers are i5 processor, 8GB RAM, and virtualisation technology enabled, Windows 10 operating system, good internet speed, and double monitors. The lab also comprises separate discussion areas for groups to interact, clear doubts and network among themselves. Google Developers Code Lab is the result of collaboration between the Centre for Advanced Studies and Google Asia Pacific Pvt. Ltd. This lab is sponsored by Google and is first of its kind in the country. As a result of the collaboration, Google organises numerous courses with well-defined curriculum, expected learning outcomes, and guidelines for lab exercises, using the lab facility for the students, faculty and staff of the University. The lab also facilitates learning through development of advanced technologies, friendly environment for discussions among peers, networking among the research groups of the University. 

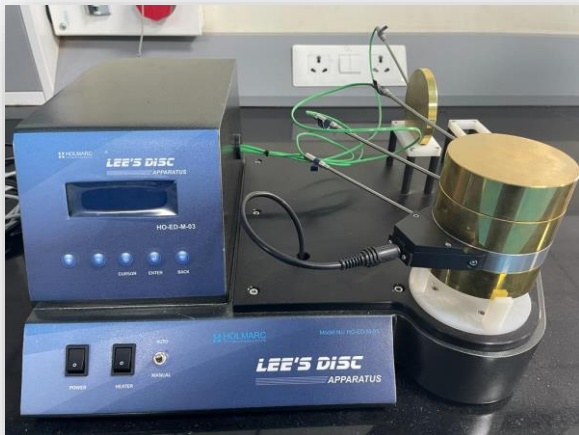
24	Augmented Reality Lab (Data Imaging & Processing Unit) HP Z2 G5 Tower Workstation and (AR Processing Unit) HPE Proliant ML350 Gen10 Server in Augmented Reality Lab	Augmented Reality Lab at Centre for Advanced Studies consists of (Data Imaging & Processing Unit) HP Z2 G5 Tower Workstation and (AR Processing Unit) HPE Proliant ML350 Gen10 Server with software solutions AR Vuforia Studio Academic SAAS, CREO Illustrate and ThingWorx Academic Teaching pack. . The objectives of this lab is to create rich 3D technical illustrations, 2D drawings, and interactive animated sequences that accurately reflect current product configurations and support formats from hard copy to augmented reality, repurpose existing 3D CAD data and animated sequences to build intuitive assembly, inspection, service and operating instructions. Efficiently integrate sensor and contextualized IoT data from the Thing Worx platform and enterprise system data. It will set the foundation for an augmented reality strategy. 
25	MICROWAVE REACTOR (ANTON-PAAR) Model No. MONOWAVE 450	The Anton Paar Monowave product line is a series of high-performance monomode microwave reactors designed for small- to medium-scale microwave synthesis. Attain temperatures up to 300 °C and pressures up to 30 bar with advanced all-the-time safety features for high-speed, high-pressure, high-temperature reactions. Improve productivity and product purity across all applications in research and development laboratories with precise heating profiles. Reduce consumable costs and our environmental footprint by using the reusable vials, caps, and septa Key Features: 24 sequential reactions, unattended, The Autosampler MAS 24 allows 24 unattended reactions, in sequence – saving time, and increasing efficiency. The factor in synthesis: Precise, scalable temperature measurement Current scientific practice is to use reaction temperature as the relevant parameter in microwave reactions, accurate measurement of the reaction temperature is the key to correct transferal and scale-up of reaction protocols between different reactors. Vials for heating any solvent and any reaction mixture In addition to the standard glass vials, silicon carbide (SiC) vessels are available. They absorb microwave energy efficiently, while shielding the interior from microwave radiation. Can choose from reaction vials for any scale between 0.5 mL and 20 mL. For bulky samples and extraction applications, a special wide-neck vessel is provided convenience without compromising performance. Watch your synthesis and analyze it in real-time: can record images and videos with the integrated digital camera.

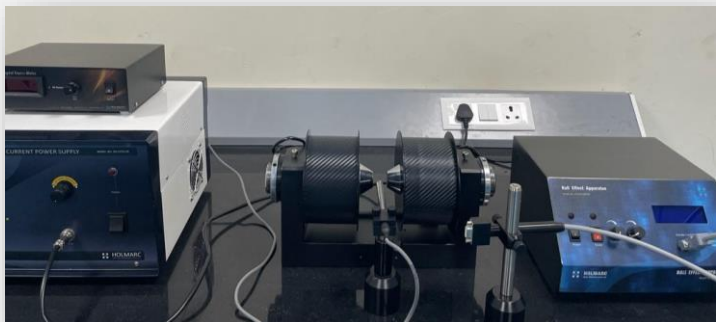
		
26	Drop Shape Analyzer – DSA100	Drop Shape Analyzer – DSA25 is high-quality system solution for almost all tasks in the analysis of wetting and adhesion on solid surfaces. DSA25 can be employed from the basic unit for precise measurement of the contact angle to expert instrument for serial measurement of surface free energy (SFE). Instrument comes with a sophisticated sample positioning systems with up to three axes and is capable of measuring contact angle. Contact angle between a liquid and a solid and Surface free energy from contact angles of several test liquids using all common models can be measured with this instrument. Following are the tasks and applications which can be performed using the DSA25 • Characterization of surface pre-treatment processes • Checking the wettability of plastic, glass, ceramic, wood or metal • Quality control for wafers and microelectronics • Checking surface cleanliness • Wetting of microscopically small surfaces 
27	Automatic Micro Hardness Testing System	Automatic Hardness Testing System-It is the standard method for measuring the hardness of metals, particularly those with extremely hard surfaces: the surface is subjected to a standard pressure for a standard length of time by means of a pyramid-shaped diamond. Procedure: To perform the Vickers test, the specimen is placed on an anvil that has a screw threaded base. 1. The indenter is pressed into the sample by an accurately controlled test force. 2. The force is maintained for a specific dwell time, normally 10 -15 seconds.

		3. After the dwell time is complete, the indenter is removed leaving an indent in the sample that appears square shaped on the surface. 4. The size of the indent is determined optically by measuring the two diagonals of the square indent. 5. The Vickers hardness number is a function of the test force divided by the surface area of the indent. The average of the two diagonals is used in the following formula to calculate the Vickers hardness. The operation of applying and removing the load is controlled automatically. 
28	Thermogravimetric analyser (TGA)	The Thermogravimetric Analyzer (TGA) is an essential laboratory tool used for material characterization. TGA is used as a technique to characterize materials used in various environmental, food, pharmaceutical, and petrochemical applications. Thermogravimetric Analysis is a technique in which the mass of a substance is monitored as a function of temperature or time as the sample specimen is subjected to a controlled temperature program (upto 1600 °C) in a controlled atmosphere of nitrogen and oxygen as per the requirement. 
29	Spray Pyrolysis Unit	Spray Pyrolysis system has been used for research laboratories in thin films, especially for solar cell development. The system automates various fatigue and error creating processes involved in the technique when performed manually. Parameters like dispensing rate of the solution and speed of spray head movement which are difficult to control manually are controlled precisely by PC based automation. A positive displacement pump controlled by stepper motor and microprocessor is used to dispense solution as per requirement. The spray head movement is also controlled by stepper motor driven linear stages in X and Y direction. The temperature of the substrate heater plate is controlled independently through a dedicated controller.

		
30	Nano fiber electrospinning	It is used to make nano and micro fibers ranging from 50nm to 5 microns in diameter. Many kind of polymers like Protein nanofibers, carbon nanotubes, inorganic nanofibers etc. can be synthesized using our nano fiber electrospinning unit. This equipment uses horizontal spinning and assures the quality of the spun fibers with no dip and uneven diameter. 
31	Four probe workstation	The four-point probe is the most commonly-used piece of equipment for measuring the sheet resistance of a material. Sheet resistance is the resistivity of a material divided by its thickness, and represents the lateral resistance through a thin square of conducting/semiconducting material. This measurement uses four probes for the measurement of conductivity and resistivity. A current is passed between the outer two probes, causing a reduction in voltage between the inner two probes. By measuring this change in voltage, the sheet resistance can then be calculated. Positioning each of the four probes individually along XYZ axes manually, while observing the sample surface under an optical microscope. Heating of the samples can also be done via thermal chuck which ranges upto 200 °C.

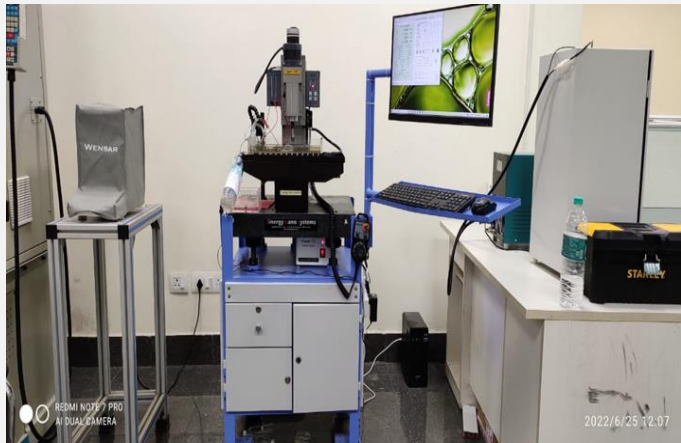
		
32	Dip Coating Unit	In Dip Coating Unit, the substrate is dipped in the solution and then withdrawn into a heating chamber. Temperature of the chamber can be controlled and fixed up to 75°C from ambient. The system has a user friendly front panel with key board and LCD display. It can as well be controlled through a computer. Dip duration, dip speed, baking duration, withdrawal speed etc. are programable features. Servo motor is used as actuator for vibration free dipping and withdrawal. A number of substrate can be held to the holder at the same time. Height of the substrate holder can be adjusted with respect to the level of the solution during the experiment. 
33	Spin Coater	Spin Coater is a dedicated tabletop system to spin coat small substrates with well controlled spin process parameters. The high speed and duration range allows the user to achieve the desired thickness or thinness of the film. The spin head actuator is a precision DC servo motor, which requires less maintenance, with accurate speed and acceleration control. A vacuum chuck powered by oil-less vacuum pump holds the substrate at the spinning head.

		The device has user friendly front panel having keyboard and LCD for programming the spin process. Spin duration, spin speed, acceleration, etc. are all programmable parameters through the front panel. As the program memory is non-volatile, the recorded parameters are not lost in case of any power failure. This model is equipped with a memory of 9 pre-set, editable programs, each having 9 steps. 
34	Lee's Disc Thermal Conductivity	Lee's Disc Apparatus is designed for the measurement of thermal conductivity in bad conductors. Thermal conductivity is the property of a material that indicates its ability to conduct heat. Conduction takes place if there exists a temperature gradient in a solid (or stationary fluid) medium. This Apparatus comprises of a brass disc resting on another slab of the same dimension with special heating coil. Both metallic discs have radial holes for the sensitive thermistors. Material under test is placed in between two discs. The heater is turned on and the apparatus is left idle until the temperature gets stabilized. At this point the heat energy passing through the heat sample will be exactly equal to the heat flowing out of the lower block. 

35	Hall Effect Measurement	Holmarc's Hall Effect apparatus is designed with state of the art modules and components. Digital display is used for all value read outs. Electromagnets, gauss meter, power supply, etc. are designed and made as separate modules for students to understand the apparatus and the principles involved easily. Safety is given due consideration in the design of the apparatus. The system consists of two cartridges, each of which is equipped with 'p' and 'n' doped germanium crystal .The cartridges can be plugged easily and safely into the D connector system. The Hall Effect set up provides all operating parameters for the samples and displays the Hall voltage, sample current as well as the sample temperature. The doped Germanium samples are used to measure the Hall-voltage as a function of the sample current, the magnetic flux density and the sample temperature. 
36	Stereo Zoom Microscope	Stereo zoom microscopes provide a stereo image when looking through the microscope. The zoom feature on the stereo zoom microscope provides a range of viewing possibilities. The zoom knob on a stereo microscope provides magnification within a large range (0.63x-6.3x) and ultimately allows the ability to view every single magnification (3.15x-378x.) when zooming in on the microscopy sample. Stereo zoom microscopes allow quick zoom from low to high magnification. Using a stereo zoom microscope makes it simple to view a sample in the entire field of view, then focus and zoom in on a particular part of interest.

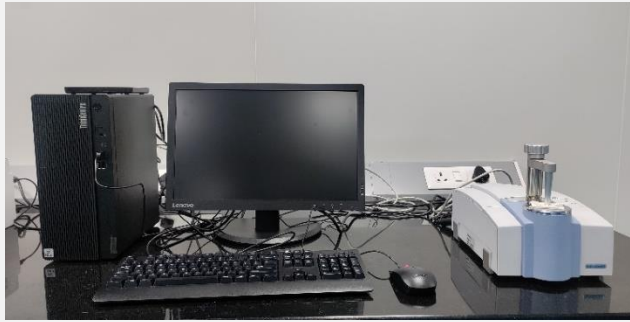
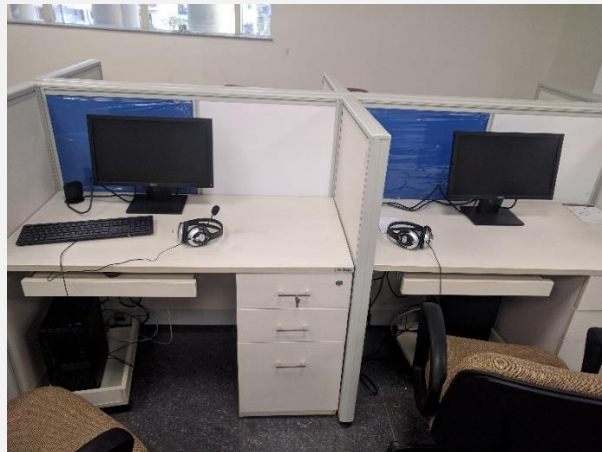
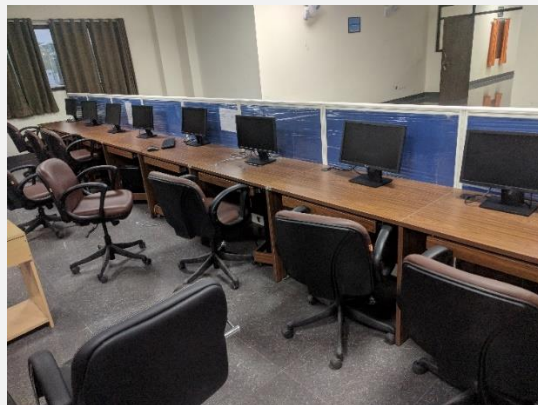
		
37	Quantum efficiency	The spectral responsivity or quantum efficiency (QE) is essential for understanding current generation, recombination, and diffusion mechanisms in photovoltaic devices. PV cell and module calibrations often require a spectral correction factor that uses the QE. The quantum efficiency in units of electron - hole pairs collected per incident photon is computed from the measured spectral response in units of amperes per watt as a function of wavelength. This system is integrated with 50W Tungsten Halogen lamp with housing, optics, power supply and an easy to use software with capability to measure the spectral response and quantum efficiency of solar cell at the wavelengths from 400nm to 1200nm. 

38	UV-Vis spectrophotometer 3092	Ultraviolet-visible (UV-Vis) spectroscopy is one of the most popular analytical techniques because it is very versatile and able to detect nearly every molecule. UV-Vis can be used in a qualitative manner, to identify functional groups or confirm the identity of a compound by matching the absorbance spectrum. UV-Vis spectroscopy is used to quantify the amount of DNA or protein in a sample, for water analysis, and as a detector for many types of chromatography. Kinetics of chemical reactions are also measured with UV-Vis spectroscopy by taking repeated UV-Vis measurements over time. 
39	Automatic Micro Hardness Testing System	Automatic Hardness Testing System-It is the standard method for measuring the hardness of metals, particularly those with extremely hard surfaces: the surface is subjected to a standard pressure for a standard-length of time by means of a pyramid-shaped diamond. Procedure: To perform the Vickers test, the specimen is placed on an anvil that has a screw threaded base. 1. The indenter is pressed into the sample by an accurately controlled test force. 2. The force is maintained for a specific dwell time, normally 10 -15 seconds. 3. After the dwell time is complete, the indenter is removed leaving an indent in the sample that appears square shaped on the surface. 4. The size of the indent is determined optically by measuring the two diagonals of the square indent. 5. The Vickers hardness number is a function of the test force divided by the Surface area of the indent. The average of the two diagonals is used in the following formula to calculate the Vickers hardness. The operation of applying and removing the load is controlled automatically.

		
40	Micro EDM Machine	Micro EDM Machine- Recent demands for micro parts have required us to manufacture variety of micro components used in different fields from entertainment electronics to biomedical implants. Recent demands for micro parts have required us to manufacture variety of micro components used in different fields from entertainment electronics to biomedical implants. Students, academicians and scientists can learn the techniques for fabrication of 3D structures on the micrometer scale. The design and construction of tools, tool holders, cutting tools, and electrodes need to evolve as greater demands are placed on them for machining these miniature parts. Miniaturization technologies are perceived as potentially key technologies of the future that will bring about completely different ways people and machines interact with the physical world. In the industrial world the interest in microscopic scale manufacturing is exponentially increasing in relation to the rapid growth of Micro Electro Mechanical Systems (MEMS) research. 

41	Optical Microscope-	Optical Microscope- Metallurgical microscopes is used analyze the surfacial properties of any material such as micro structure, grain boundry and porosity etc. It can also used for metallography, forensic, processing plastics, inspection of metals, semiconductor materials, laser systems, solar cells using both thin films and wafer technology and many other fields of material science. Students, academician and researchers can analyze and observe glossy metal surfaces that include metallurgy, mineralogy, and gemology. 
42	Die Electric-Discharge Machining (with Mql System)	Die Electric-Discharge Machining (with Mql System)-Electrical discharge machining (EDM) is a non-traditional machining process based on removing material from a part by means of a series of repeated electrical discharges between tools, called electrodes, and the part being machined in the presence of a dielectric fluid. Students, academician and scientists can learn how to machine the difficult to machine and hard to machine components. 
43	PLC SCADA-Part-A	PLC SCADA-Part-A- This lab has various types of PLC along with interfacing modules to demonstrate the detailed applications of PLC and SCADA. The purpose of this laboratory is to train the students to be familiar with the software and hardware of PLC so that they can gain enough experiences to meet the demand of the automation era. The facilities in the laboratory enable students to build a firm background in PLC hardware as well as software. Students learn about ladder logic programming, wiring different I/O's (analog and digital) with PLC and SCADA programming. They acquire the practical skills sufficient to design and realize basic automation process.

		
44	Atomic Force Microscopy NX10	The Instrument Park NX10 is today's standard AFM for materials research. With Park NX10, materials scientists can demystify the nanoscale fundamentals of innovative materials. Researchers can secure a high rate of productivity and accuracy when scanning a material regardless of the sample's softness, especially when using True Non-Contact mode. 
45	FTIR spectrophotometer Alpha -II	FTIR spectroscopy is used to quickly and definitively identify compounds such as compounded plastics, blends, fillers, paints, rubbers, coatings, resins, and adhesives. It can be applied across all phases of the product lifecycle including design, manufacture, and failure analysis. This makes it a useful tool for scientists and engineers involved in product development, quality control, and problem solving. Key areas where infrared analysis adds value include: ➤ Material identification and verification ➤ Copolymer and blend assessment ➤ Additive identification and quantification ➤ Contaminant identification—bulk and surface ➤ Molecular degradation assessment

		
46	Language Lab with headphones and i7 Systems	The language lab provides access to native-speakers via audio-video aids so that they learn correctly. Language lab would help students in studying the language of the country where they are planning to pursue their education. As strong communication skills are essential in almost all of the professional careers, language lab can help in acquiring this important skill. Language laboratory at Centre of Advanced Studies consists of dell optilex system with intel i7 processors, headphones with mic and audio visual softwares. 
47	Linux/ Open Source Lab	The Linux/ Open Source lab provides support for the educational and research interests of students and faculty in high level computing, high performance and distributed systems, and open source computing. CAS vision is to develop such high-performance, practical solutions, and contribute to the needs of both scientific and engineering community. The lab is equipped with a 22 dell optilex systems with intel i7 processors running on ubuntu linux and kali linux os. 

48	General Purpose Computing Lab	This lab is for general uses, basic and advanced programming Masters students and contains 32 dell optilex systems .All the machines are running Windows 10 , 64-bit, with standard development software packages for lab based teaching and coursework installed. List of softwares include : Application Softwares: Adobe Reader, Firefox, Google Chrome, Flash Player, Vlc Player, Skype, Office, Photoshop, Notepad++, MikTex, Sublime, Virtual Box, SqlServer System Software: Java Sdk, Power Shell, GNS3, Android sdk, Andrid Emulator, Turbo C++, Wamp Server, PyChem, Visual Studio, CodeBlock. 
49	Drone Systems Lab	Laboratory is focused on the design, engineering, and implementation of multimodal imaging systems, calibration techniques, and processing algorithms to provide the highest quality aerial data. The lab has three drone systems for this purpose. A professional grade drone system with a maximum takeoff weight is 4.25 kg including gimbal and full HD camera with an approximate flight time of 27 min. A mini drone system with max take-off weight of 249g with HD camera, with maximum flight time of 31 min and a DIY drone kit. An industry standard multispectral sensor which can simultaneously capture up to six bands with 2cm resolution when flying at 60m that can be attached with professional drone system for capturing multispectral image data.   Professional Drone System  Mini Drone System  Multi-Spectral Camera  DIY Drone Kit